

Proximal Point Algorithm

$$\min: f(x)$$

$$\text{prox}_{\lambda f}(\hat{x}) = \underset{x}{\operatorname{argmin}}: f(x) + \frac{1}{2\lambda} \|x - \hat{x}\|^2$$

$$\text{Prox Point Alg (PPA)}: x_{t+1} = \text{prox}_{\lambda f}(x_t)$$

When is this a good idea?

What are properties?

Proximal Point Algorithm - continued -

PPA applied to a linear approximation of f
gives GD

$$f^{(1)}(x) = f(x_t) + \langle \nabla f(x_t), x - x_t \rangle$$

$$x_{t+1} = \text{prox}_{\lambda f^{(1)}}(x_t) = x_t - \lambda \nabla f(x_t) \quad (\text{check})$$

PPA applied to a 2nd order approx of f :

$$f^{(2)}(x) = f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + \frac{1}{2} (x - x_t)^T \nabla^2 f(x_t) (x - x_t)$$

$$x_{t+1} = \text{prox}_{\lambda f^{(2)}}(x_t) = x_t - \left(\nabla^2 f(x_t) + \frac{1}{\lambda} I \right)^{-1} \nabla f(x_t)$$

¹ Modified - Hessian - Newton, or Levenberg-Marquardt algorithm

Resolvent of Monotone Operator

Recall that f is convex $\Leftrightarrow y_1 \in \partial f(x_1), y_2 \in \partial f(x_2)$

$$\Leftrightarrow \langle y_1 - y_2, x_1 - x_2 \rangle \geq 0$$

∂f is monotone when f is convex.

We can generalize this idea. $\partial f: \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ i.e., $\partial f(x) \subseteq \mathbb{R}^n$

$T: \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ is called monotone if $y_i \in T(x_i) \Rightarrow \langle y_i - y_j, x_i - x_j \rangle \geq 0$

Defn: The Resolvent of operator T is: $\boxed{(I + \lambda T)^{-1}}$

Proposition: $\text{prox}_{\lambda f}(\hat{x}) = (I + \lambda \partial f)^{-1}(\hat{x})$

pf: $x \in (I + \lambda \partial f)^{-1}(\hat{x}) \Leftrightarrow \boxed{\hat{x} \in (I + \lambda \partial f)(x)} = \boxed{x + \lambda \partial f(x)}$

$$\Leftrightarrow 0 \in \partial f(x) + \frac{1}{\lambda}(x - \hat{x}) \Leftrightarrow x \in \underset{u}{\text{argmin}} : f(u) + \frac{1}{2\lambda} \|u - \hat{x}\|_2^2$$



Proximal Point Algorithm vs Gradient Descent

$$\text{PPA: } x_{t+1} = \text{prox}_{\lambda f}(x_t)$$

$$\Leftrightarrow x_{t+1} = (I + \lambda \partial f)^{-1}(x_t)$$

$$\Leftrightarrow x_t \in (I + \lambda \partial f)(x_{t+1}) = x_{t+1} + \lambda \partial f(x_{t+1})$$

$$x_{t+1} \in x_t - \lambda \partial f(x_{t+1})$$

$$\text{If } \partial f(x_{t+1}) = \nabla f(x_{t+1})$$

$$\Rightarrow \boxed{x_{t+1} = x_t - \lambda \nabla f(\underline{x_{t+1}})}$$

$$\left[\begin{array}{l} \text{GD:} \\ x_{t+1} = x_t - \lambda \nabla f(\underline{x_t}) \end{array} \right]$$

PPA "backwards" ^{op}, ^{Implicit algorithm} GD "forward"

Note: For small λ , $\underset{\text{PPA}}{(I + \lambda \nabla f)^{-1}} \approx \underset{\text{GD}}{I - \lambda \nabla f}$

Proximal Point in the Dual

$$(P) \quad \min_x: f(x) + g(Ax)$$

$$(D) \quad \max_z: \underbrace{-f^*(-A^T z) - g^*(z)}_h$$

$$z_{t+1} = \text{prox}_{\lambda h}(z_t) = \arg \min_z: f^*(-A^T z) + g^*(z) + \frac{1}{2\lambda} \|z - z_t\|_2^2$$

$$\text{opt. conditions: } 0 \in -A \partial f^*(-A^T z_{t+1}) + \partial g^*(z_{t+1}) + \frac{1}{\lambda} \underbrace{(z_{t+1} - z_t)}_{\downarrow}$$

$$\Rightarrow \boxed{z_{t+1} = z_t + \lambda (A \hat{u} - \hat{y})}$$

$$\text{where: } \hat{u} \in \partial f^*(-A^T z_{t+1}), \quad \hat{y} \in \partial g^*(z_{t+1})$$

Compare to GD in the dual (Gradient Ascent)

$$\min: f(x) + g(Ax)$$

$$\max: -f^*(-A^T b) - g^*(z)$$

$$\text{GA: } z_{t+1} = z_t + \lambda \boxed{w} \quad \hookrightarrow \in \partial(-f^*(-A^T z_t) - g^*(z_t))$$

$$w \in A \partial f^*(-A^T z_t) - \partial g^*(z_t)$$

$$w = A \hat{z} - \hat{y} \quad \text{where: } \hat{z} \in \partial f^*(-A^T z_t), \hat{y} \in \partial g^*(z_t)$$

$$\hat{z} \in \arg\min: f(x) + z_t^T A x$$

$$\hat{y} \in \arg\min: g(y) - z_t^T y$$

PPA in the Dual: Augmented Lagrangian

$$\max_x: -f^*(-A^T z) - g^*(z)$$

$$\text{PPA: } z_{t+1} = \text{prox}_{\lambda h}(z_t) = \arg \min_z: f^*(-A^T z) + g^*(z) + \frac{1}{2\lambda} \|z - z_t\|^2$$

$$\Leftrightarrow z_{t+1} = z_t + \lambda (A \hat{x} - \hat{y}) \quad (\star)$$

$$\hat{x} \in \partial f^*(-A^T z_{t+1}), \quad \hat{y} \in \partial g^*(z_{t+1})$$

$$\Leftrightarrow 0 \in \partial f(\hat{x}) + A^T \underline{z_{t+1}}, \quad 0 \in \partial g(\hat{y}) - \underline{z_{t+1}}$$

$$\text{plug in } (\star): z_{t+1} = z_t + \lambda (A \hat{x} - \hat{y})$$

$$\Leftrightarrow 0 \in \partial f(\hat{x}) + A^T (z_t + \lambda (A \hat{x} - \hat{y})), \quad 0 \in \partial g(\hat{y}) - (z_t + \lambda (A \hat{x} - \hat{y}))$$

$$\Leftrightarrow 0 \in \partial f(\hat{x}) + A^T z_t + \lambda A^T (A \hat{x} - \hat{y}),$$

$$\Leftrightarrow (\hat{x}, \hat{y}) \in \arg \min_{x, y}: f(x) + g(y) + z_t^T (Ax - y) + \frac{\lambda}{2} \|Ax - y\|_2^2$$
$$= \arg \min_{x, y}: f(x) + g(y) + \frac{\lambda}{2} \|Ax - y + z_t/\lambda\|_2^2$$

Augmented
Lagrangian